Mobina Pournemat

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Mountain View, California

Overview

I am a third-year Computer Science Ph.D. student at the University of Maryland, College Park, advised by Prof. Soheil Feizi. My research centers on reasoning in large language models — in particular **efficient inference and reasoning**, **reasoning faithfulness**, and the **evaluation** of what these models can do reliably. I also have experience in multimodal generative models, computer vision, and applied machine learning in industry.

Education

- **University of Maryland, College Park** — Maryland, USA
  Ph.D. in Computer Science; Advised by Prof. ***Soheil Feizi****; GPA: 4/4* — *Aug. 2024 - Present*
- **Sharif University of Technology** — Tehran, Iran
  B.Sc. in Computer Engineering — *Sep. 2018 - Feb. 2023*

Research Experience

- **University of Maryland, College Park** — Maryland, USA
  Research Assistant; Advisor: Prof. Soheil Feizi — *Aug. 2024 - Present*
  - **Efficient Reasoning through Cross-Model Latent Knowledge Transfer (in progress)**:
    Reducing long chain-of-thought inference cost by transferring reasoning knowledge across models in latent space.
  - **Exploring LLMs' Probabilistic Reasoning Capabilities (EMNLP 2026, with Meta AI)**:
    Built a comprehensive benchmark for reasoning under uncertainty and uncovered systematic failure modes.
  - **Evaluating LLM Reliability Beyond Accuracy**:
    Quantified how consistently LLMs answer semantically identical questions.
  - **Fine-Grained Instruction Control for Image Editing (CVPR 2026)**:
    Contributed to SliderEdit, a continuous, per-instruction control interface for instruction-based image editing.
- **Institute of Science and Technology Austria (ISTA)** — Klosterneuburg, Austria
  Machine Learning Intern; Supervisor: Prof. Johann Danzl — *Nov. 2023 - Jul. 2024*
  - **3D Semantic Segmentation, Microscopy Image Analysis**:
    Developed an efficient pipeline for automated semantic segmentation of brain microscopy images using *Segment Anything*.
- **Sharif University of Technology** — Tehran, Iran
  NLP Research Assistant; Advisor: Dr. Ehsaneddin Asgari — *Jul. 2022 - Oct. 2023*
  - **Persian Syntactic and Semantic Error Correction Tool**:
    Developed an auto-correction tool for Persian text, combining a BERT-based language model with rule-based modules.
- **Sharif University of Technology** — Tehran, Iran
  Computer Vision Research Assistant; Advisor: Dr. Mohammad Hossein Rohban — *Sep. 2021 - Jul. 2022*
  - **Anomaly Detection with Autoencoders; Semantic Segmentation**:
    Benchmarked reconstruction- and representation-based methods on standard vision datasets.

Publications

- **Reasoning Under Uncertainty: Exploring Probabilistic Reasoning Capabilities of LLMs**
  M Pournemat, K Rezaei, G Sriramanan, A Zarei, H Eghbalzadeh, Y Wang, J Fu, F Peng, Z Wang, X Feng, N Noorshams, S Feizi - (**EMNLP 2026**, Main Conference)
- **SliderEdit: Continuous Image Editing with Fine-Grained Instruction Control**
  A Zarei, S Basu, M Pournemat, S Nag, R Rossi, S Feizi - (**CVPR 2026**, Oral Presentation)
- **Same Question, Different Answers: Evaluating LLM Reliability Beyond Accuracy**
  K Faghih, Y Cheng, S Saha, M Pournemat, A Gerami, S Feizi - (Under Review)

Work Experience

- **Behin Bazaar Marketing Agency** — Tehran, Iran
  Data Analyst, Business Intelligence — *Apr. 2023 - Oct. 2023*
  - **Data Analysis, Database Design, Data Preprocessing, Visualization, Reporting**:
    Designed the analytics database and delivered SQL and Power BI dashboards used for campaign decisions.
- **AIMedic** — Tehran, Iran
  Machine Learning for Healthcare Intern — *Jul. 2021 - Sept. 2021*
  - **Data Science, Classification, Semantic Segmentation**:
    Built and evaluated classification and segmentation models for medical imaging data.

Selected Projects

- **Quantifying Attributional Bias in LLM Toxicity Detection** (Nov. 2025):
  Measuring how toxicity scores for identical comments change when attributed to different author identities.
- **Benchmarking Multi-Stage Vulnerability Reasoning in LLMs** (Oct. 2025):
  Evaluating the effect of explicit reasoning and post-verification on LLM performance in vulnerability detection.
- **Explainable Detection of Online Sexism** (May 2022):
  SemEval-2023, Task 10: hierarchical classification and explanation of sexist content.
- **Recommender System Design** (May 2021):
  Content-based, collaborative filtering, and hybrid recommenders with offline evaluation of ranking quality.

Teaching Assistant Experience

- **Introduction to Artificial Intelligence**: Fall 2024
- **Data Structures and Algorithms**: Fall 2022
- **Natural Language Processing**: Spring 2022
- **Computer Simulation**: Fall 2021
- **Probability and Statistics**: Spring 2021

Relevant Coursework

- **Natural Language Processing**: 4/4 (A)
- **Long-Context Language Models**: 4/4 (A)
- **Multimodal Foundation Models**: 4/4 (A)
- **Large Language Models Security and Privacy**: 4/4 (A)
- **Advanced Numerical Optimization**: 4/4 (A)
- **Artificial Intelligence**: 20/20
- **Data Structures and Algorithms**: 19.6/20
- **Stanford CS229 - Machine Learning**: online, audited
- **Natural Language Processing Specialization**: Coursera

Technical Skills

- **Programming Languages:** Python, Java, C, SQL, JavaScript, PHP
- **Machine Learning Libraries:** PyTorch, Hugging Face Transformers, TensorFlow Keras, NumPy, Pandas, scikit-learn
- **Web Development & Database:** HTML/CSS, jQuery, Bootstrap, MySQL, PostgreSQL, SQL Server
- **Miscellaneous:** Microsoft Office, Power BI, Docker, Git, Linux, LaTeX

Languages

Persian: native **English:** proficient **Spanish:** low proficiency **Turkish:** low proficiency